

System Overview

Patient

- Register with personal info (name, age, gender, contact details, medical history).
- Log in securely.
- Book an appointment with a doctor.
- View appointment status.

Doctor

- Register with professional info (specialization, available times).
- Log in securely.
- View patients who booked appointments.
- Update appointment status and notes.

Main Features

Authentication & Registration

- Patients and doctors can register separately with role-based fields.
- Login system with session tracking.
- Logout function to clear session.

Appointments

- Patients can **book** an appointment with a doctor.
- Doctors can **see appointment details** for their patients.
- Appointment includes **status** (Pending, Confirmed, Completed) and **notes**.

Database Management

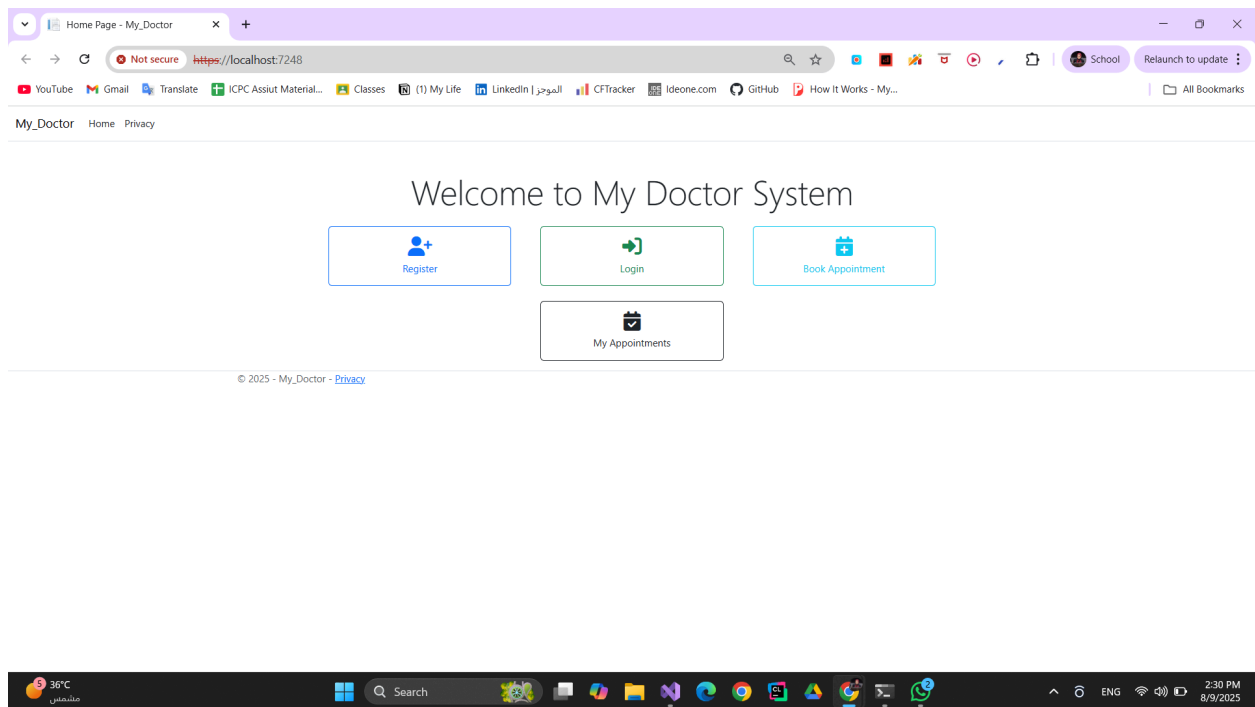
- Uses **Entity Framework** to manage:
 - **User** (base class for Doctor and Patient)
 - **Doctor**
 - **Patient**
 - **Appointment**
 - **Prescription**

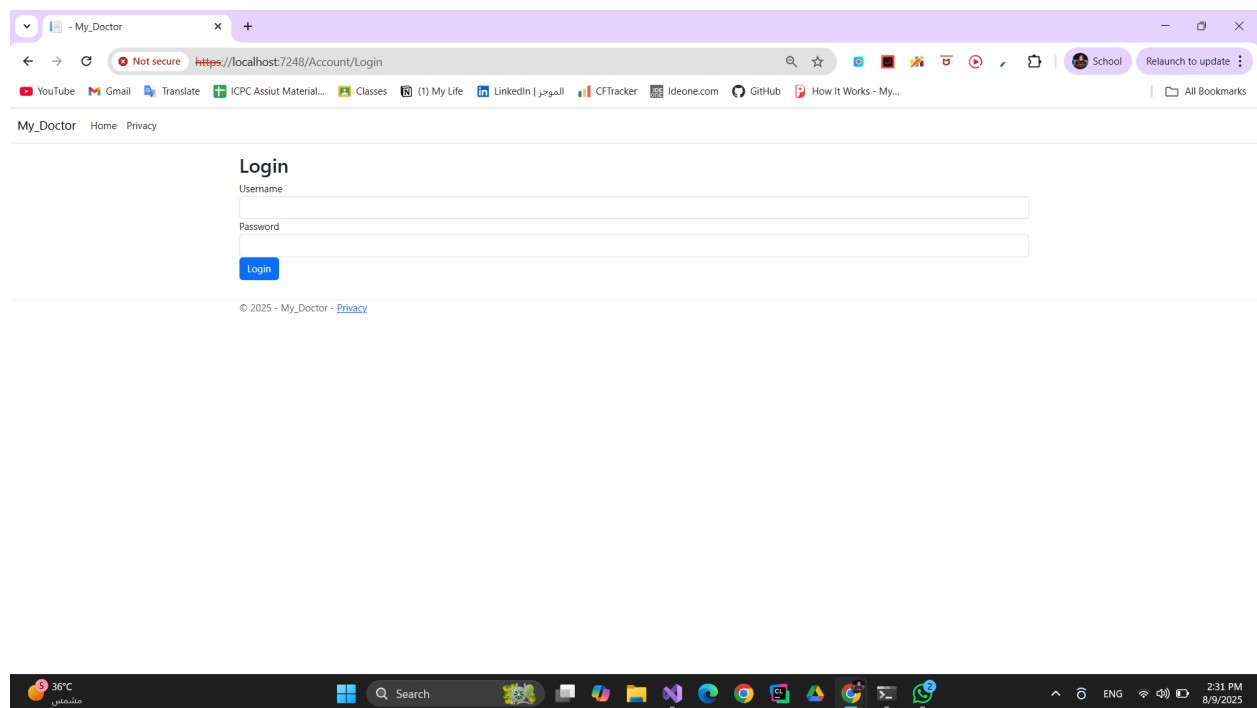
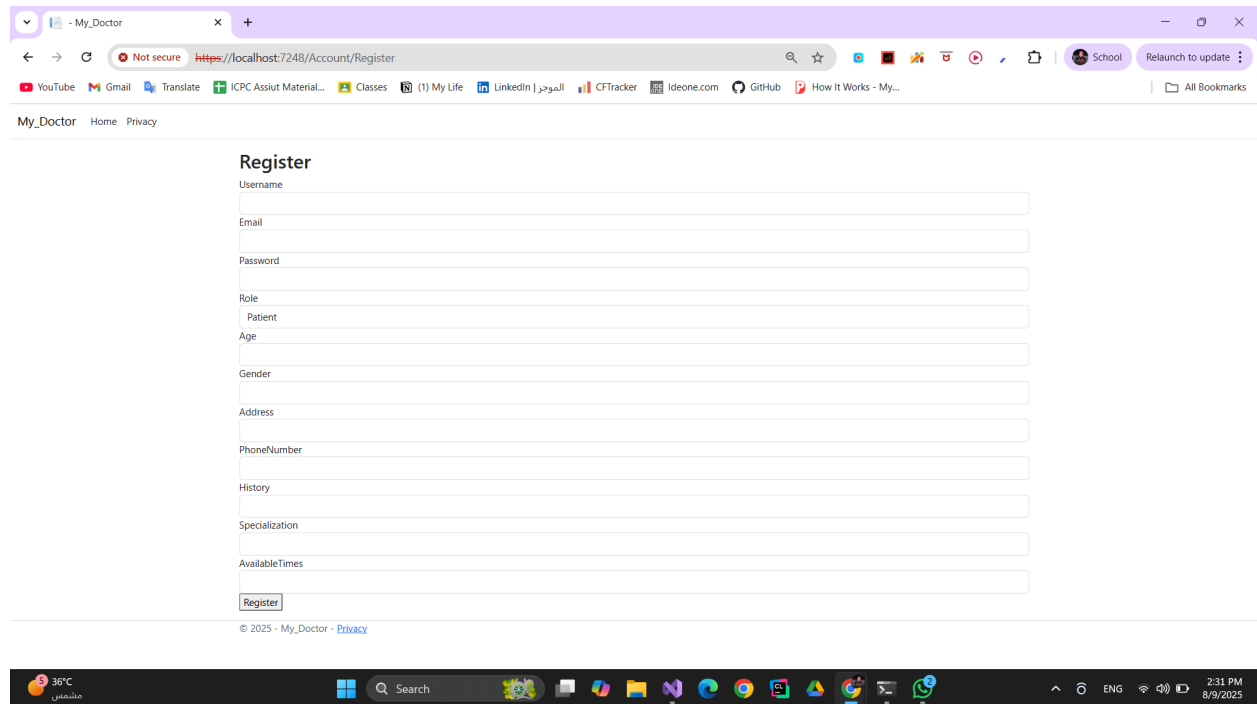
Frontend

- Razor pages for:
 - Register
 - Login
 - Book Appointment
 - View Appointment List
- Bootstrap or FontAwesome icons for navigation.

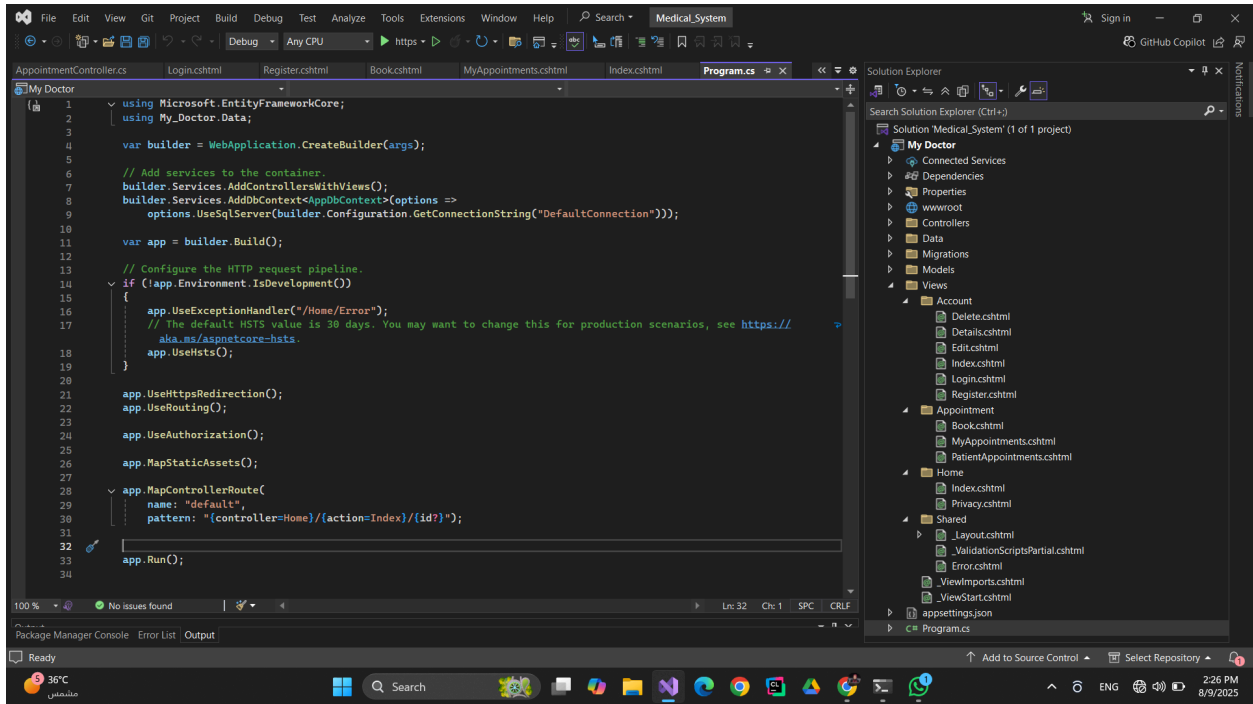
Workflow

1. **From Home Page** → click **Register** → fill patient or doctor form.
2. **Login** → enter username/password.
3. **If Patient:**
 - Book appointment with a doctor.
 - View appointment list.
4. **If Doctor:**
 - See list of patients who booked appointments.
 - Update their status & add notes.

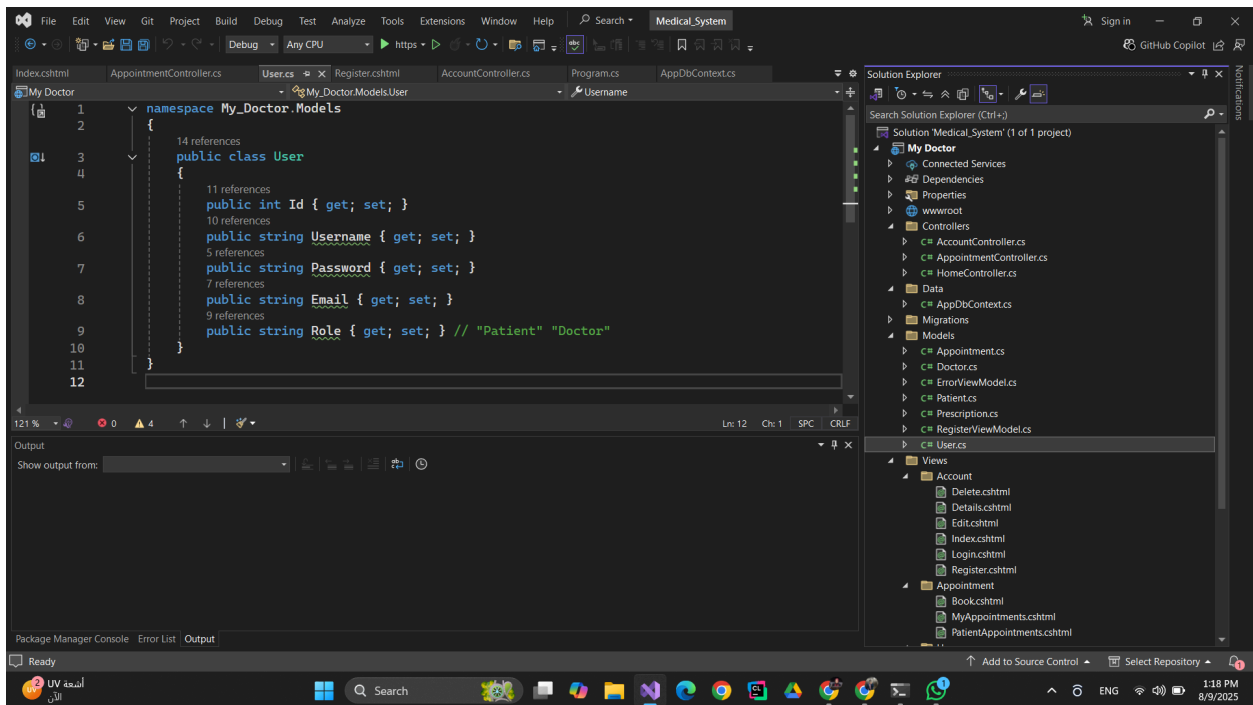


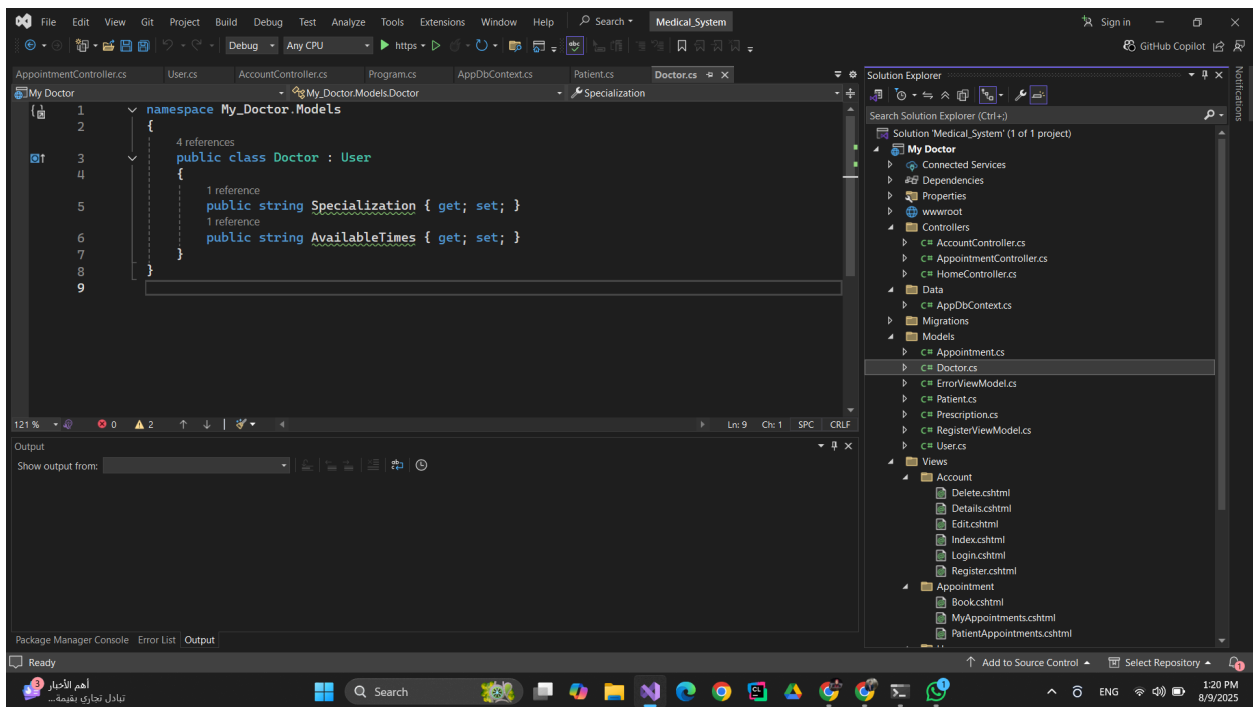
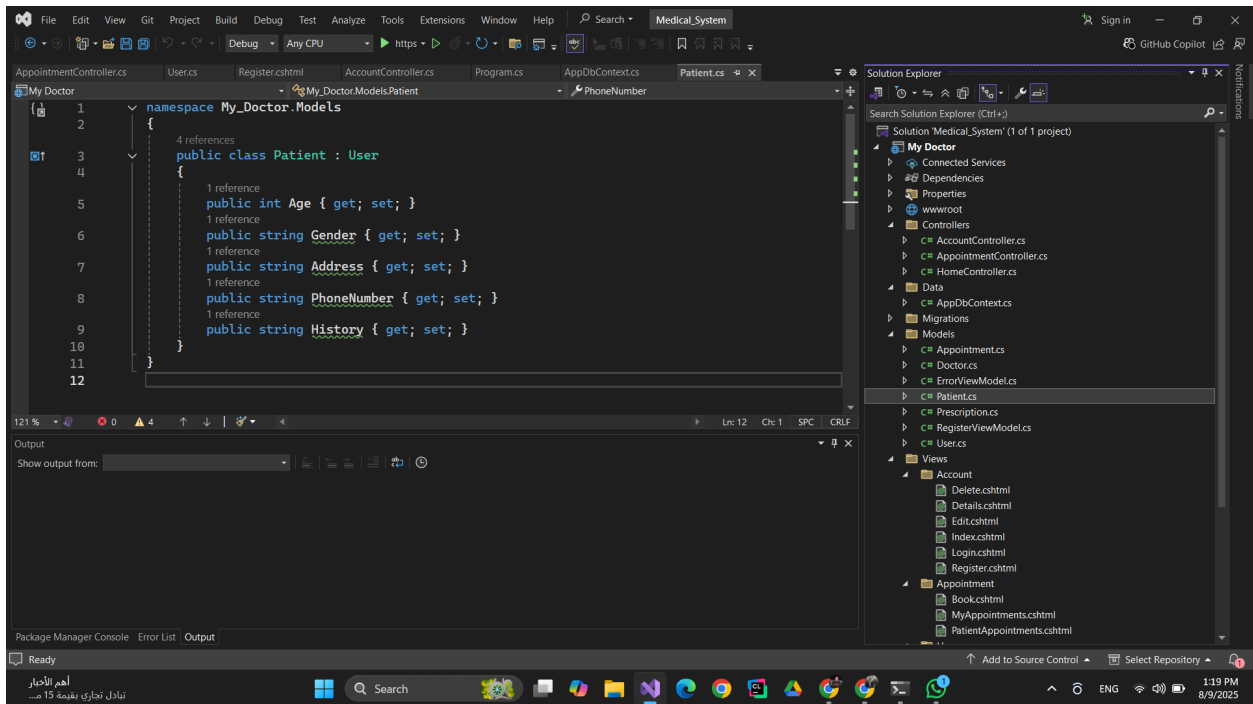


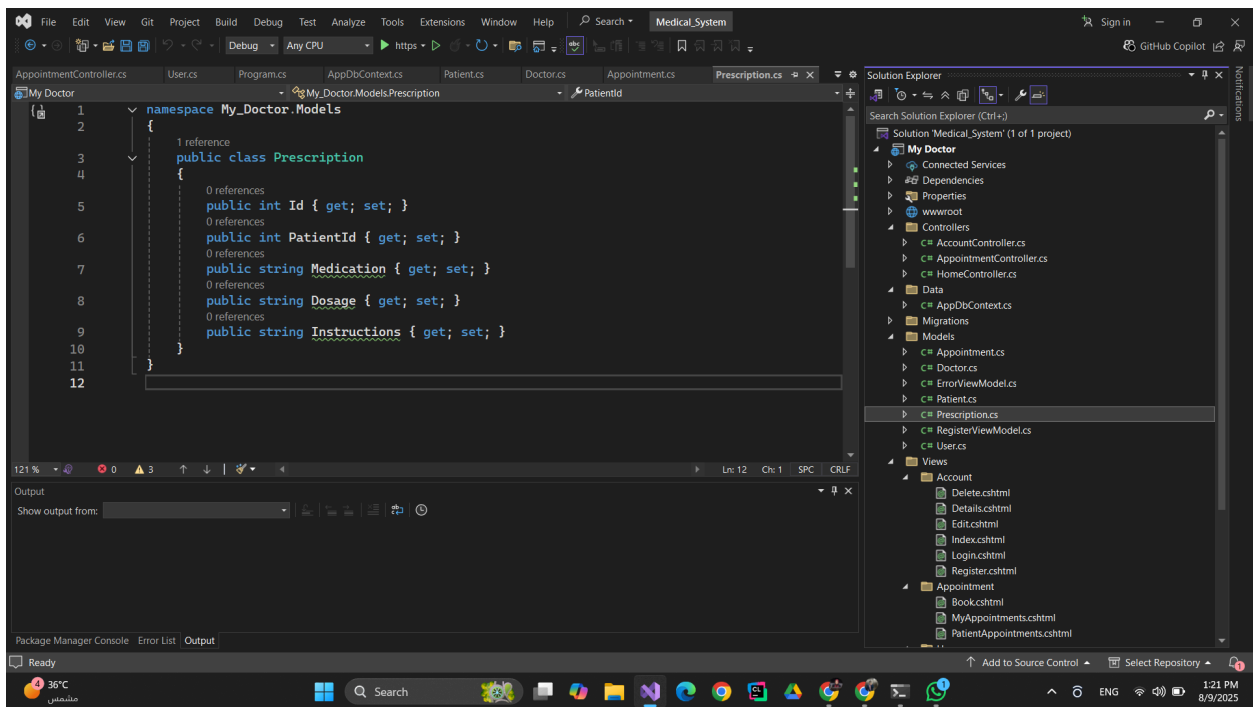
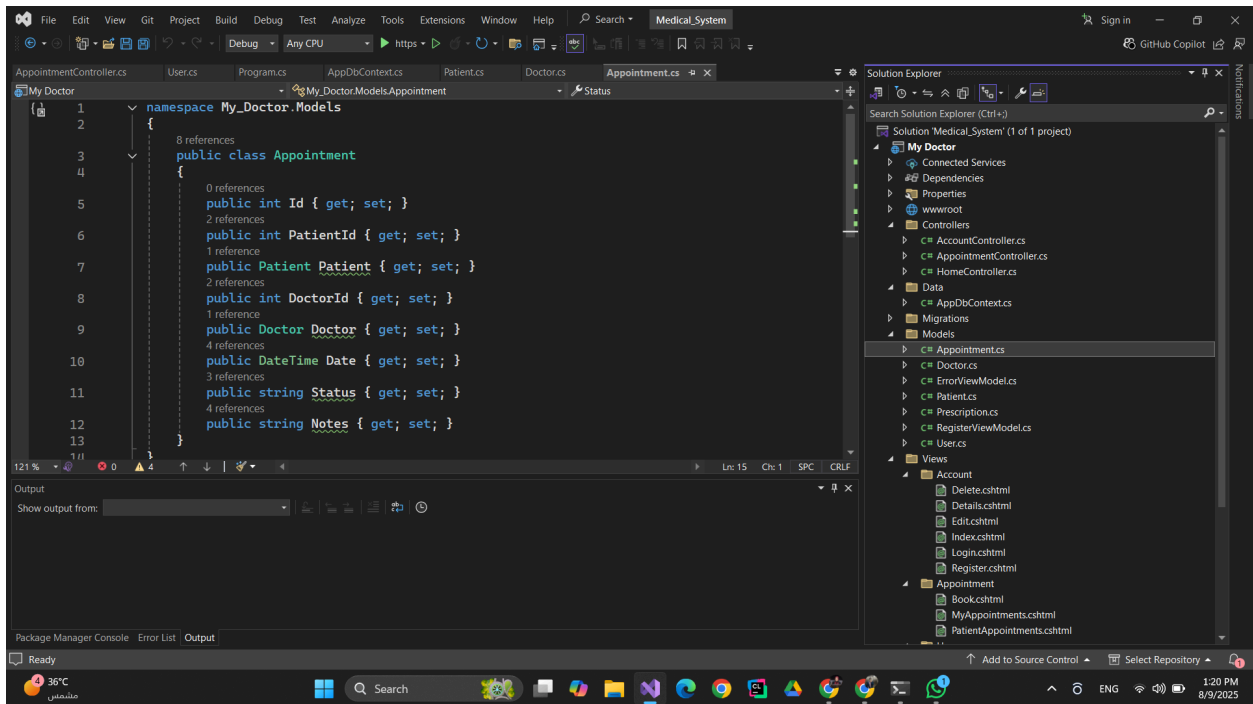
Program



Entities :







Entities (Models Folder in Project)

User (Base) : Id, Username, Password, Email, Role (Patient/Doctor)

Patient : Inherits User,+Age,Gender,Address,PhoneNumber,History

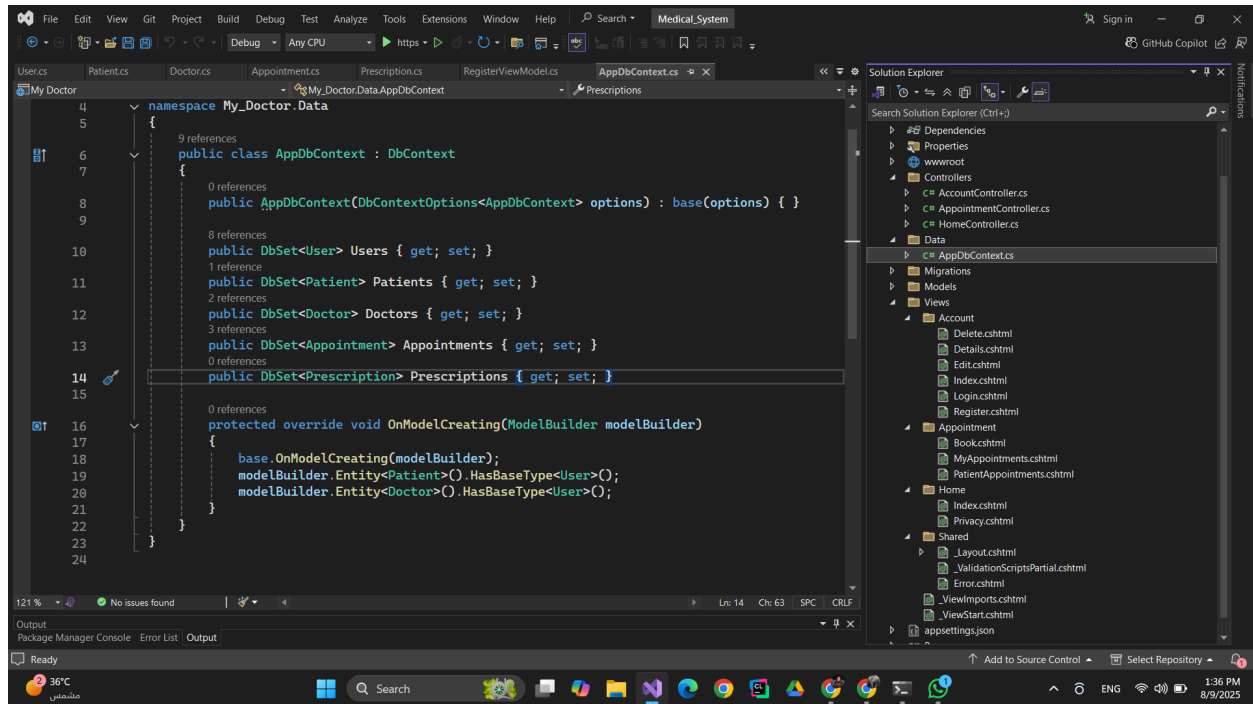
Doctor : Inherits User, + Specialization, AvailableTimes

Appointment : Id, PatientId, DoctorId, Date, Status, Notes

Prescription : Id, PatientId, Medication, Dosage, Instructions

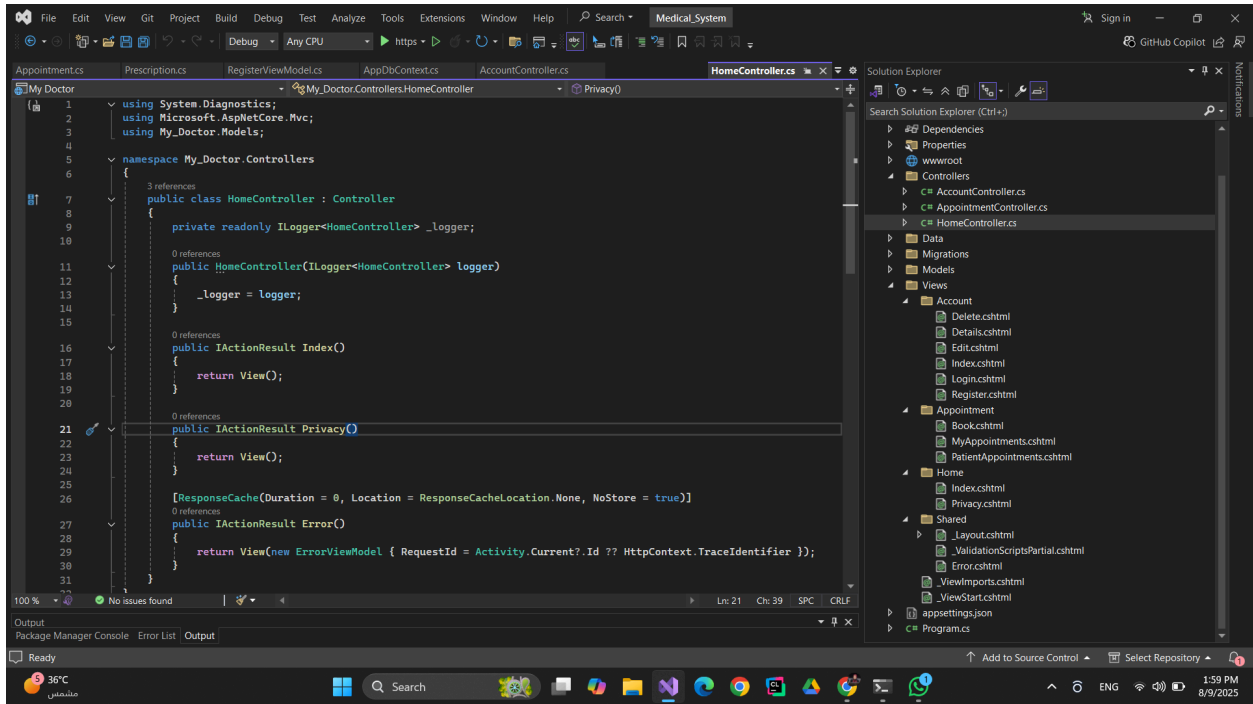
Database (Data Folder in Project)

Store Users Data in Database for the next time when the same user use the website using SQL Server



Controllers Folder :

HomeController : is the main entry point for general pages in the **My_Doctor** system.



Main Functions

1. Index()

- Loads the home page (Views/Home/Index.cshtml).

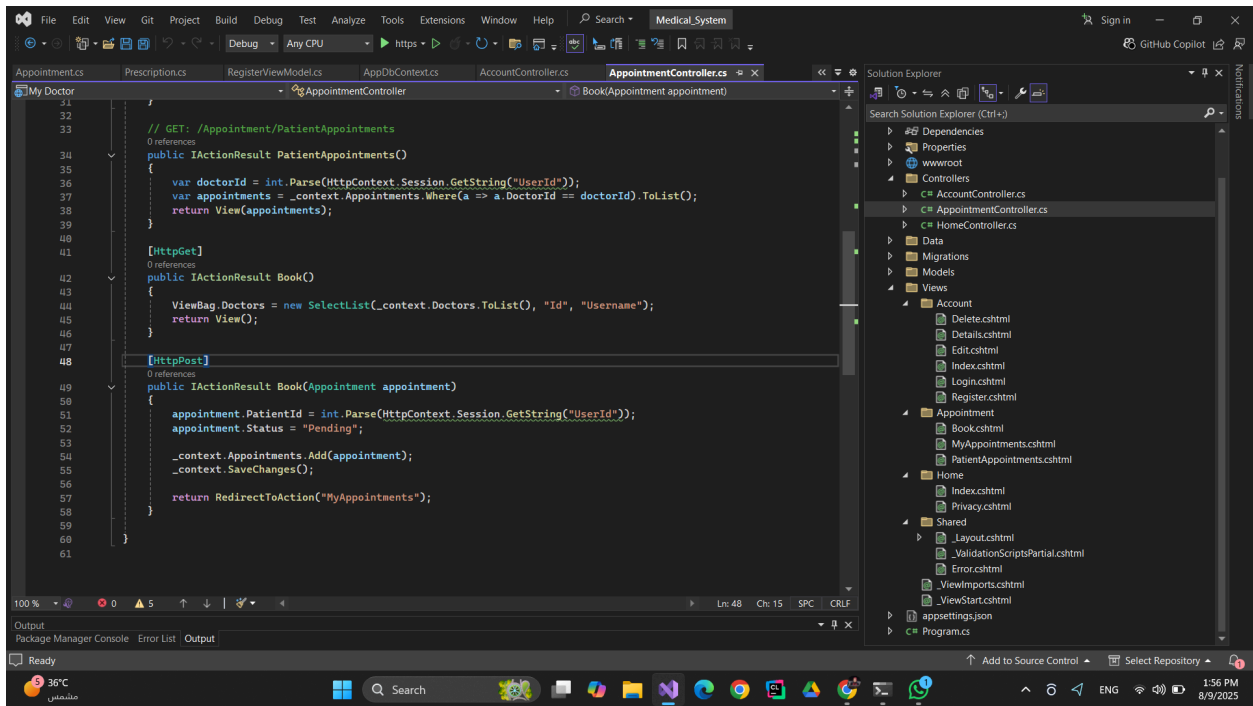
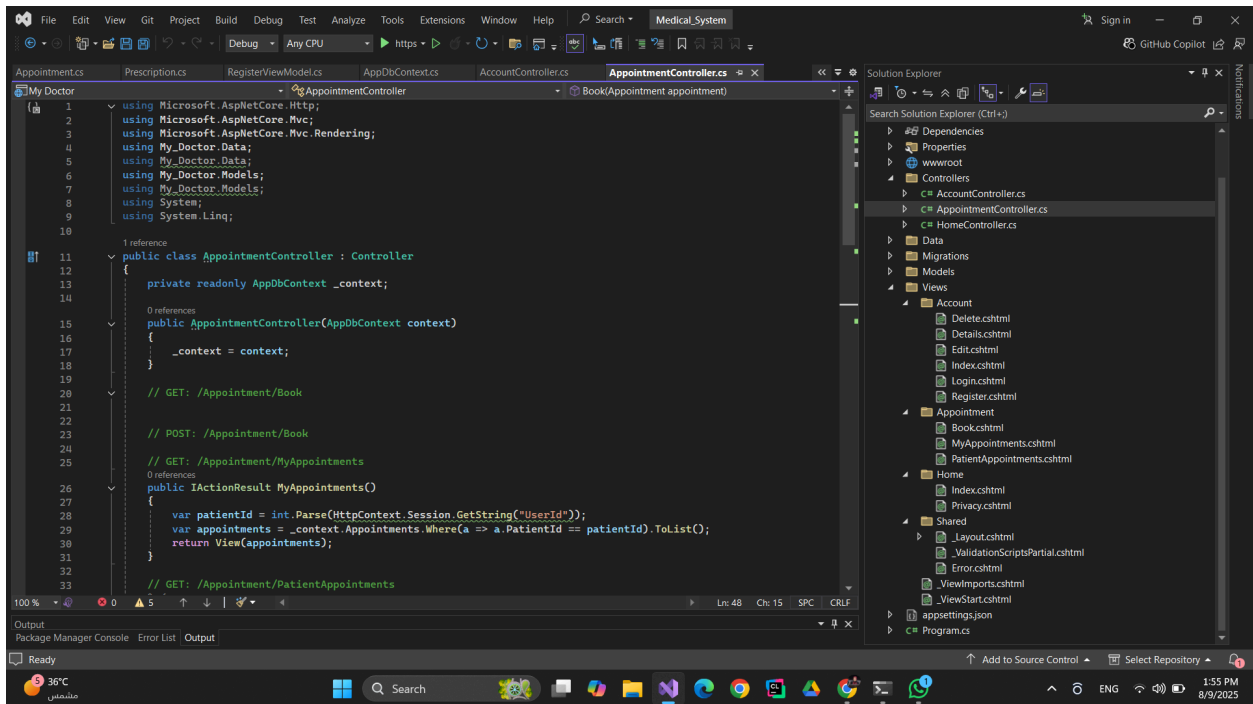
2. Privacy()

- Loads the privacy policy page (Views/Home/Privacy.cshtml).

3. Error()

- Shows the error page with a RequestId for debugging.
- This is used when something goes wrong in the system.

AppointmentController : is responsible for booking and viewing appointments between patients and doctors.



Main Functions :

1. Booking an Appointment

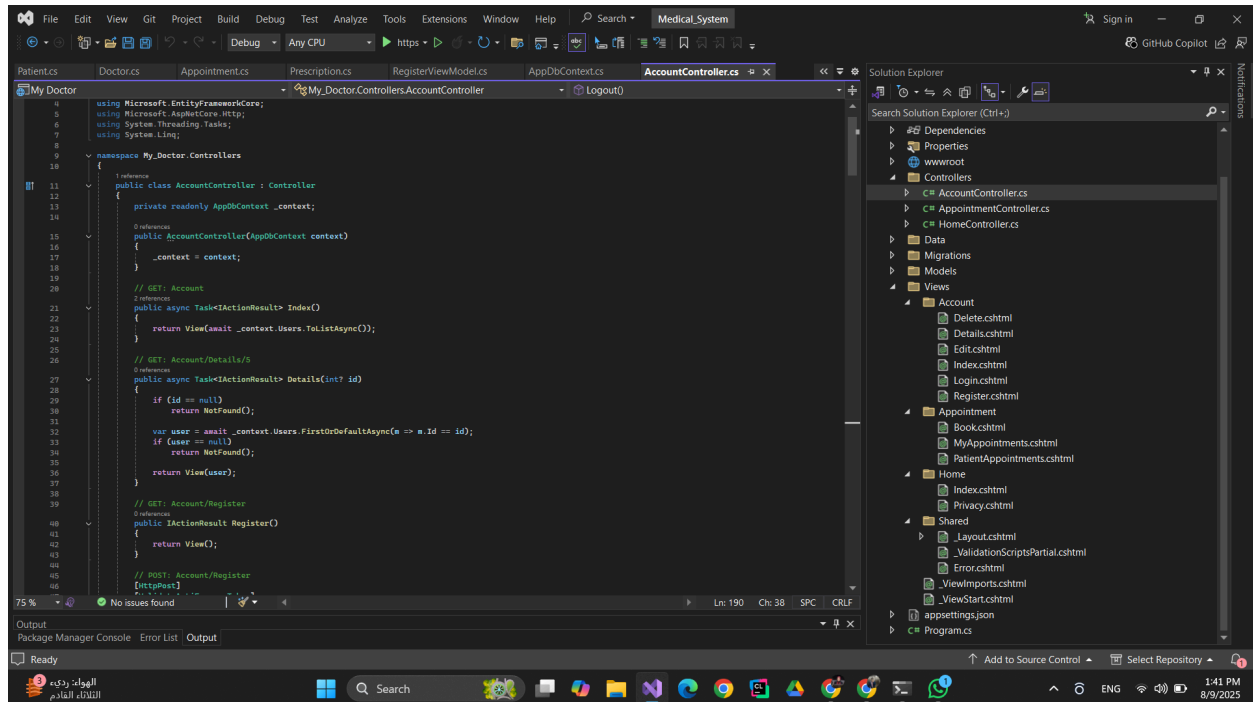
- GET: Book()

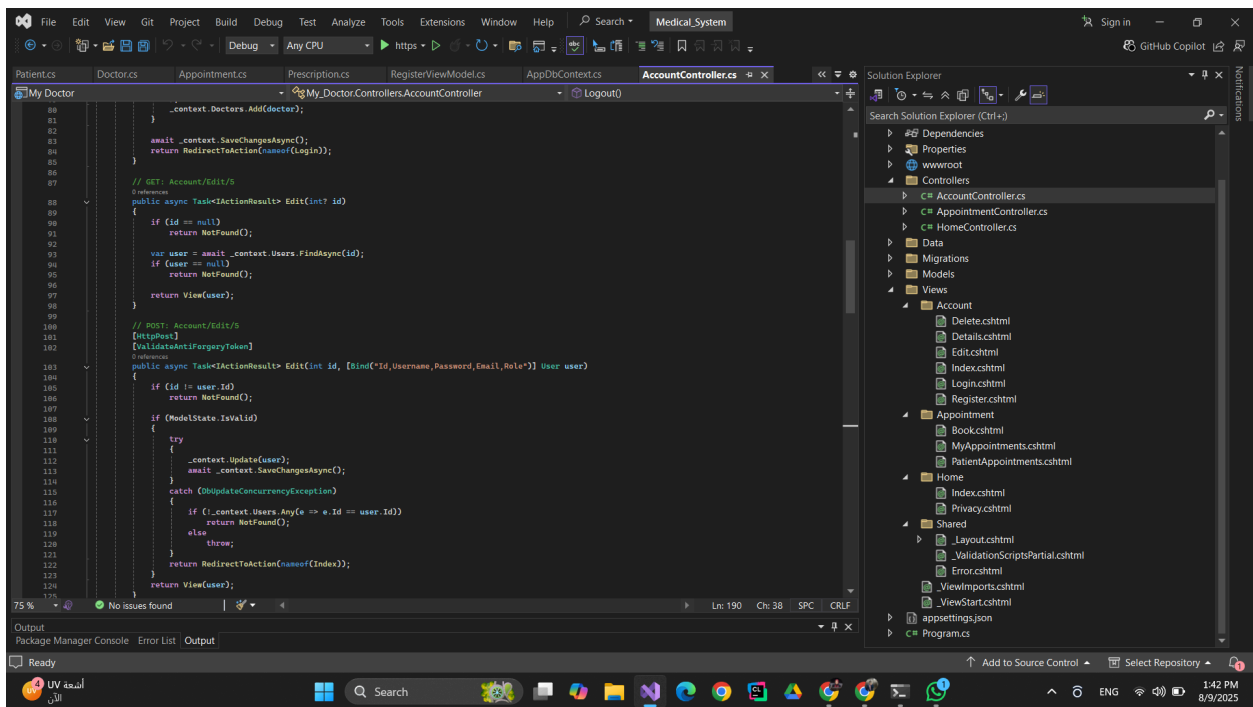
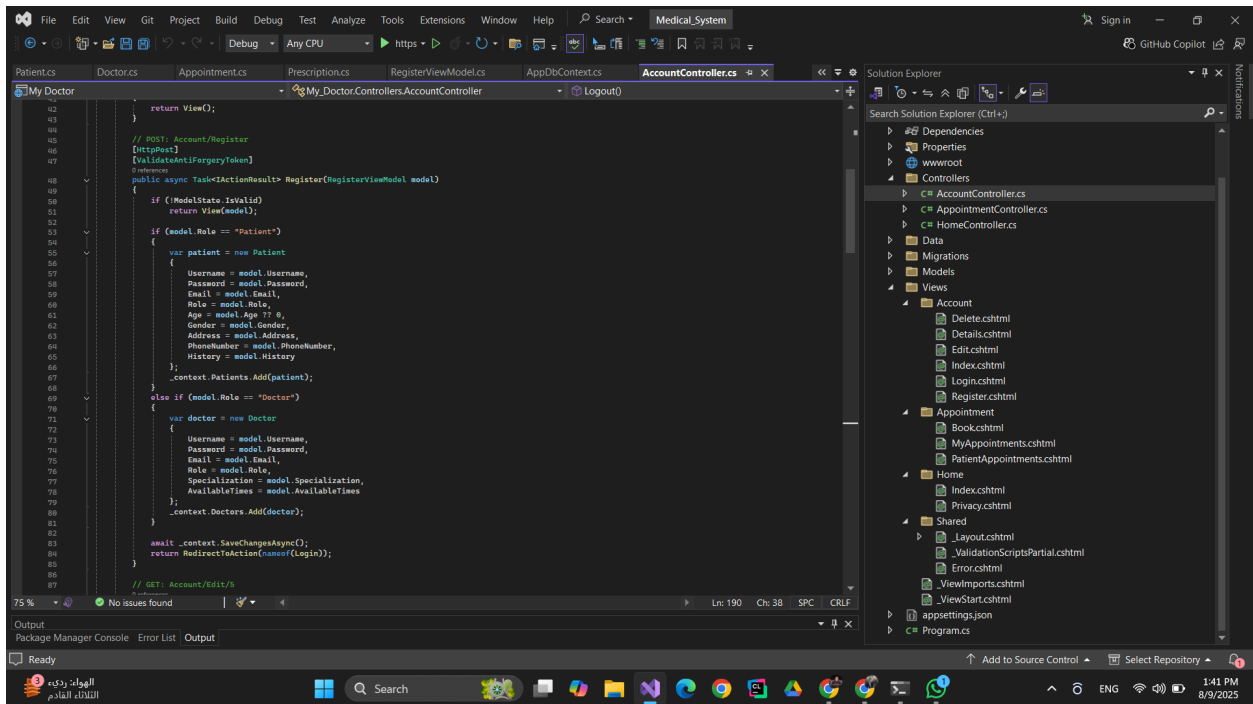
- Fetches all doctors from the database.
- Sends the doctor list to the view using **ViewBag.Doctors** so the patient can choose a doctor.
- **POST: Book(Appointment)**
 - Gets the logged-in **PatientId** from the session.
 - Sets **Status** to "Pending".
 - Saves the appointment to the database.
 - Redirects the patient to **MyAppointments**.

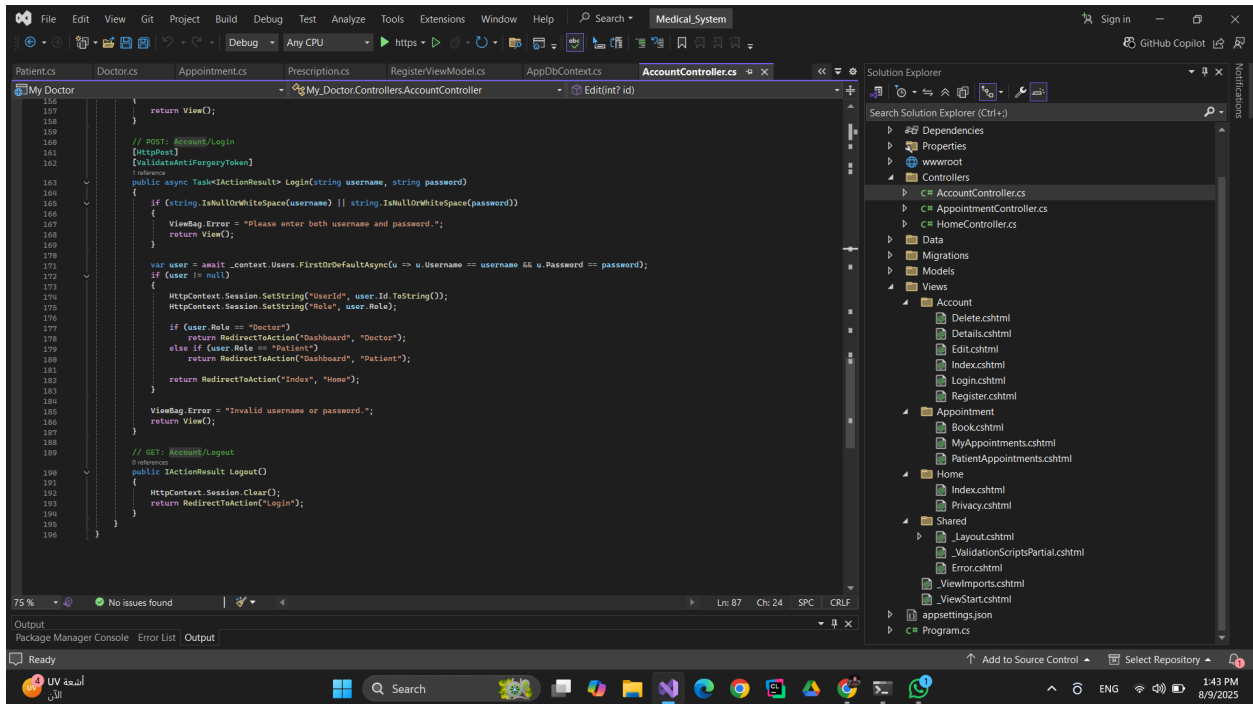
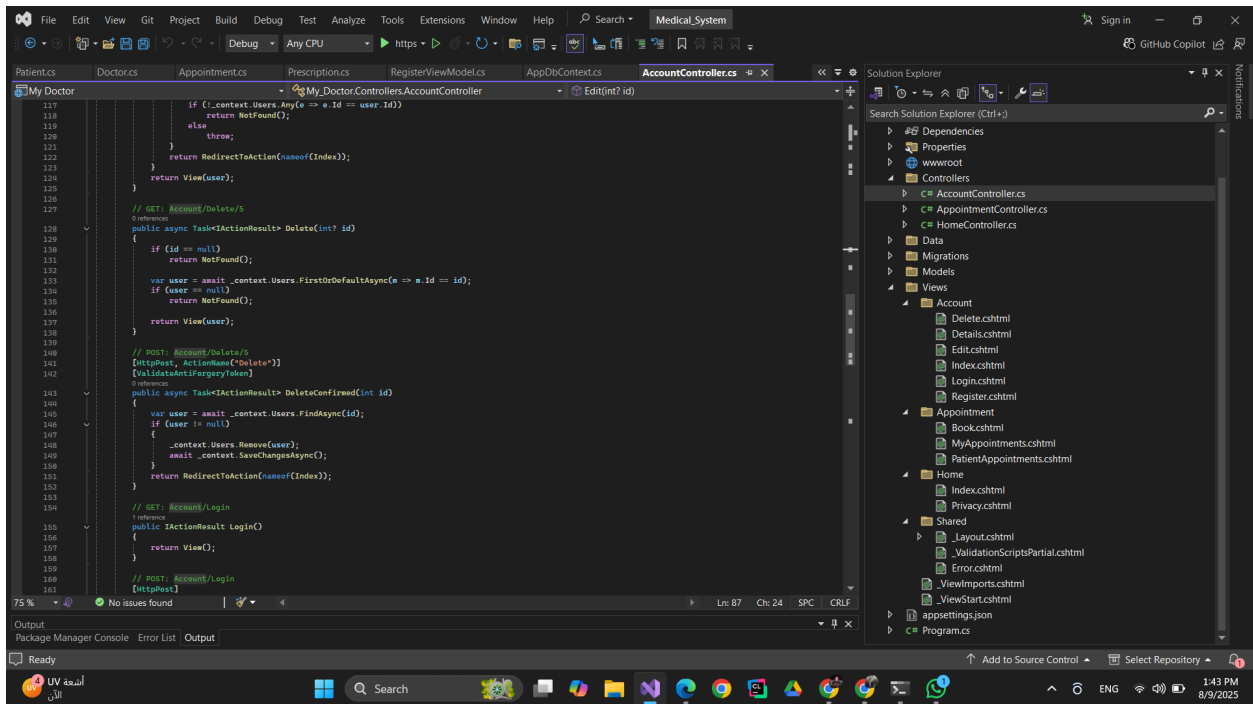
2. Viewing Appointments

- **MyAppointments()** (for Patients)
 - Finds all appointments where **PatientId** matches the logged-in patient.
 - Displays them in a list.
- **PatientAppointments()** (for Doctors)
 - Finds all appointments where **DoctorId** matches the logged-in doctor.
 - Displays them in a list.

AccountController : It handles registering users, logging in/out, and managing user profiles.







Main Functions :

1. User Management

- Index() → Lists all users in the system.
- Details(id) → Shows information for a specific user.

- **Edit(id)** → Allows updating a user's information.
- **Delete(id)** → Allows deleting a user

2. Registration

- **GET: Register()** → Displays the registration form.
- **POST: Register(model)** →
 - Checks if the data is valid.
 - If role is "**Patient**", creates a **Patient** record with personal info (age, gender, address, phone, history).
 - If role is "**Doctor**", creates a **Doctor** record with specialization and available times.
 - Saves the new user in the database.

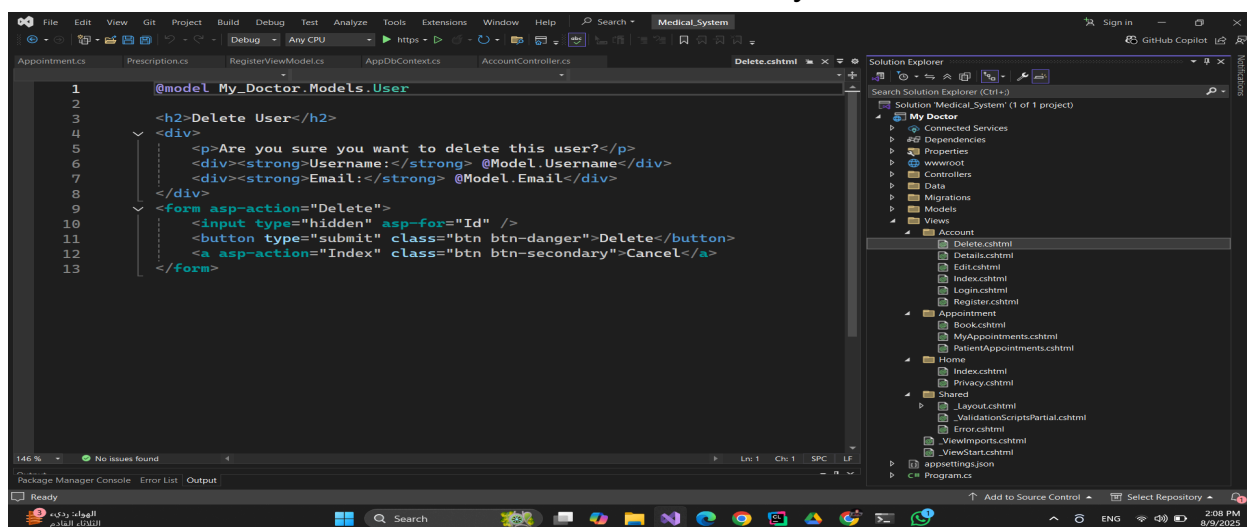
3. Login & Logout

- **GET: Login()** → Displays the login form.
- **POST: Login(username, password):**
 - Validates credentials.
 - Stores **UserId** and **Role** in the session.
 - Redirects doctors to Doctor Dashboard and patients to Patient Dashboard.
- **Logout()** → Clears session and redirects to the login page.

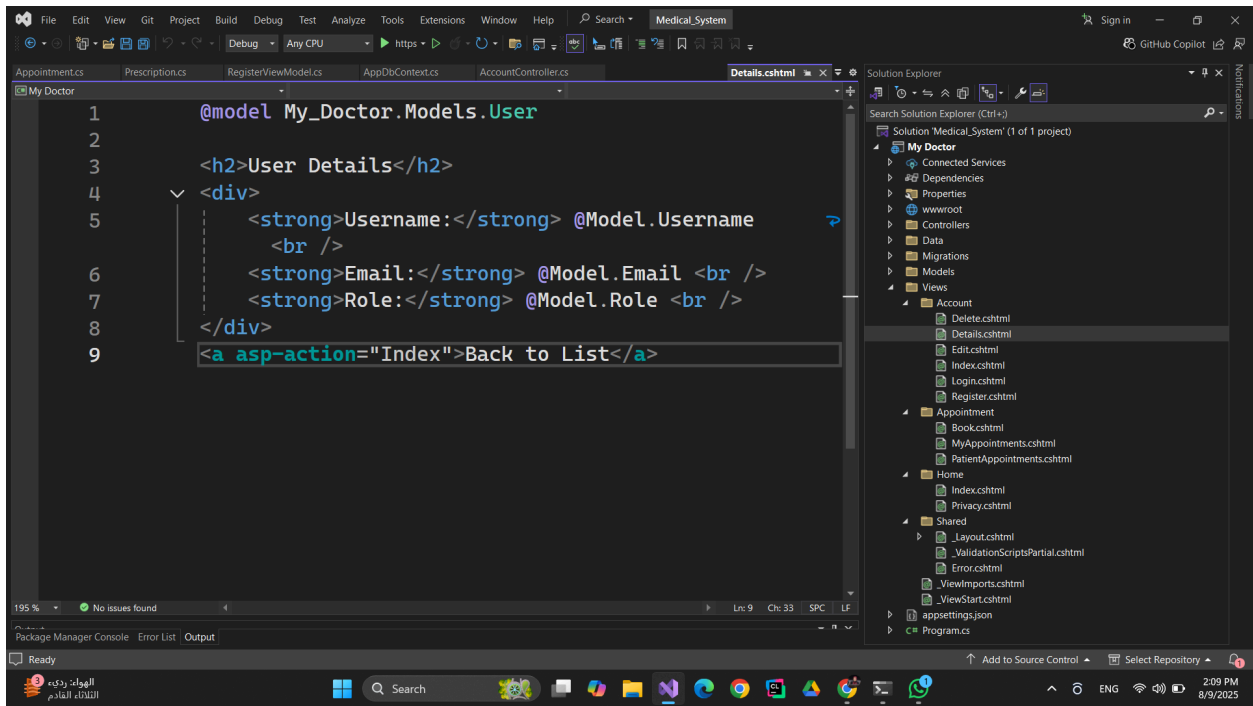
Views Folder :

Account :

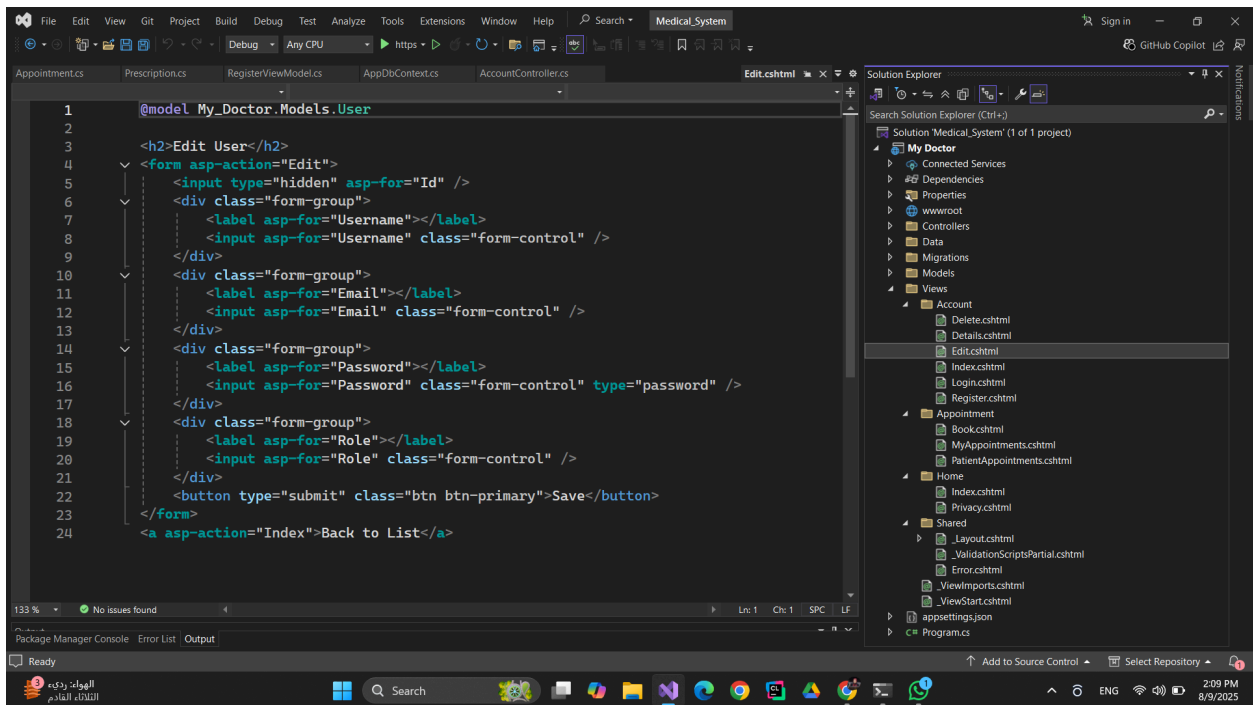
Delete : Confirms and deletes a user from the system.



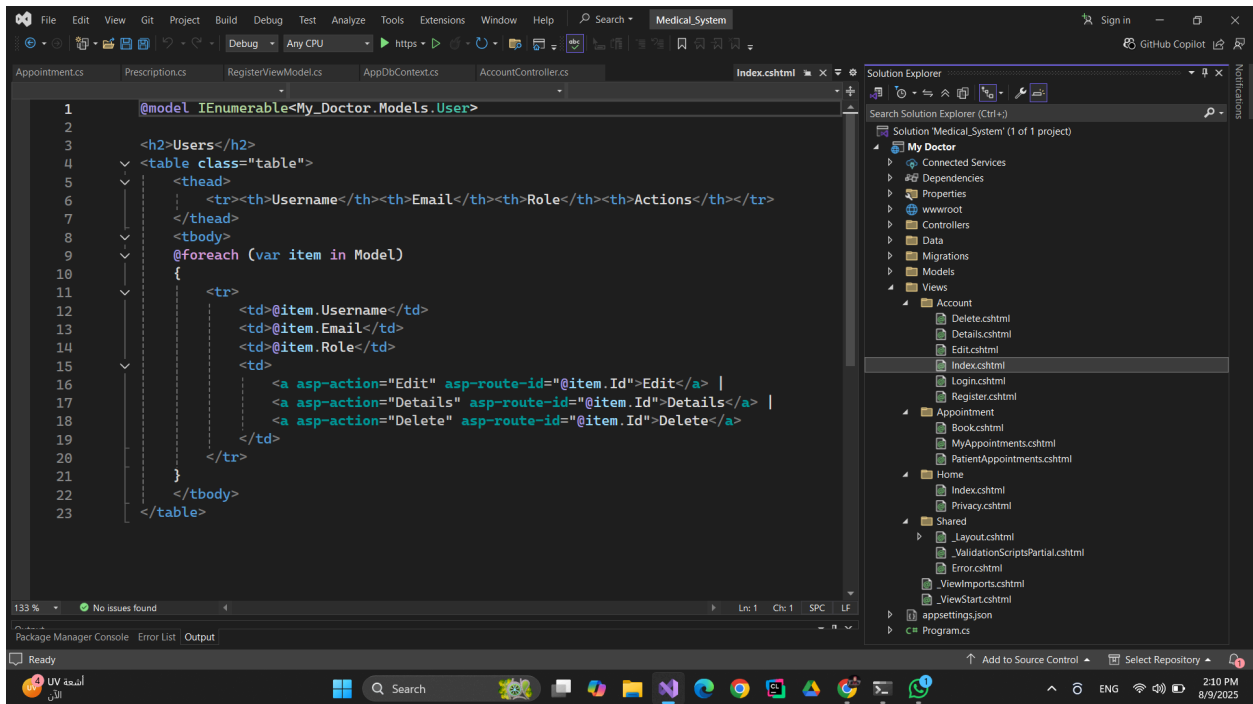
Details : Shows detailed information for a selected user.



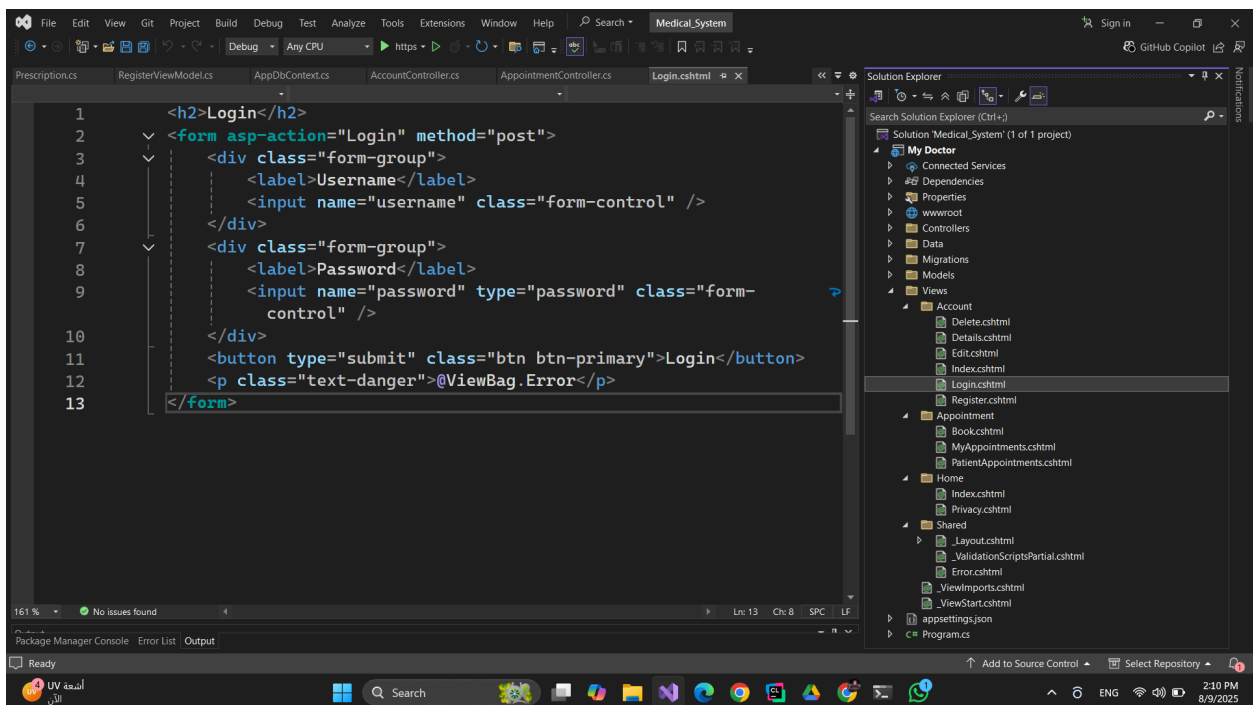
Edit : Allows updating an existing user's information.



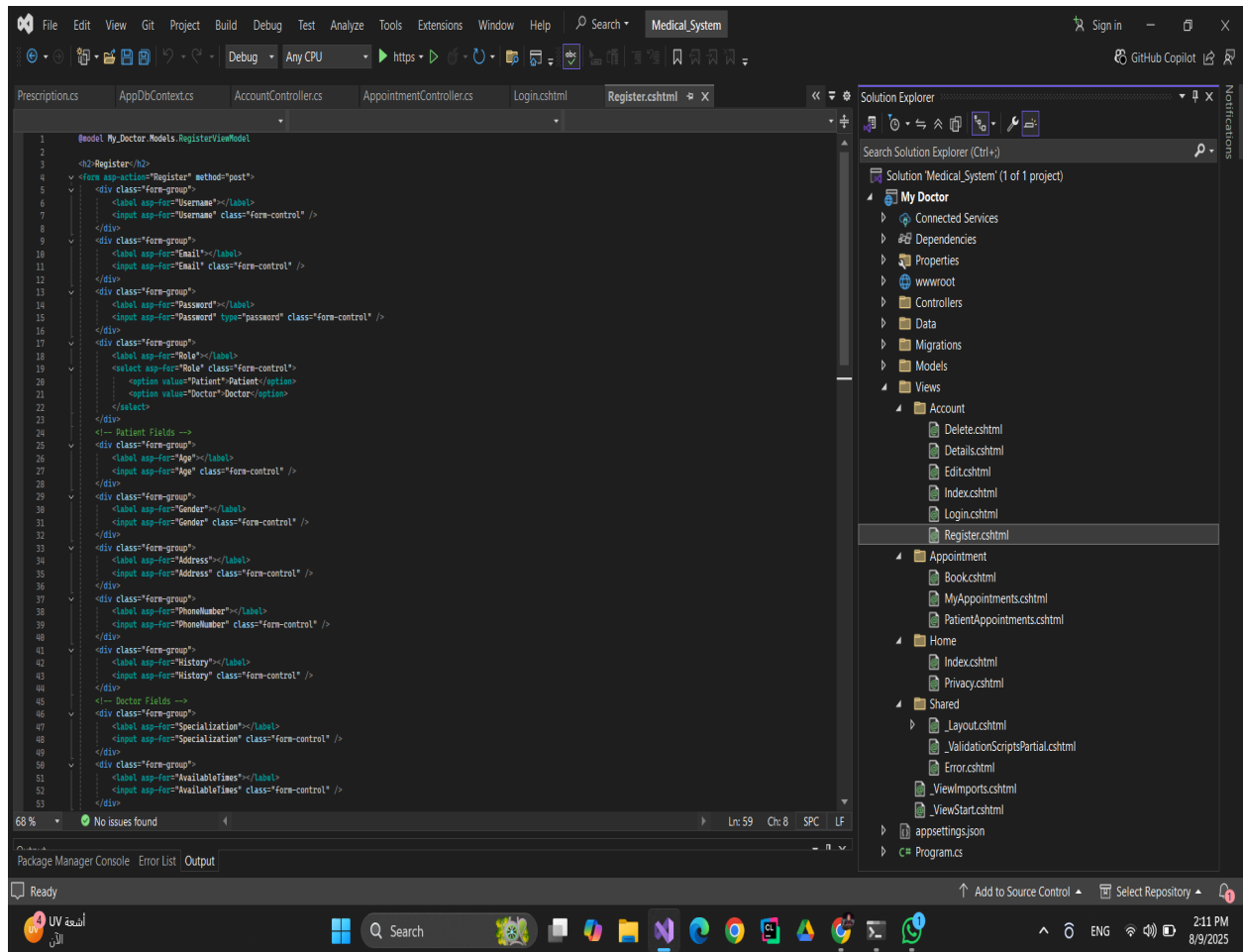
Index : Displays a list of all registered users.



Log in : Authenticates a user and starts a session.

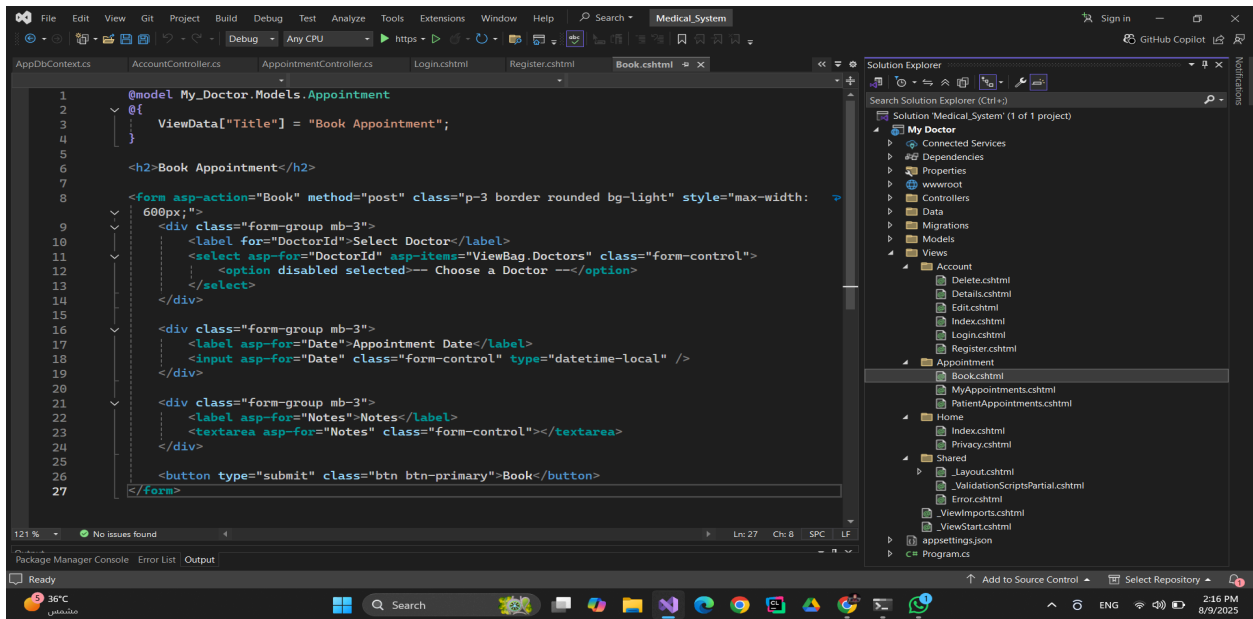


Register : Provides a form to create a new patient or doctor account.

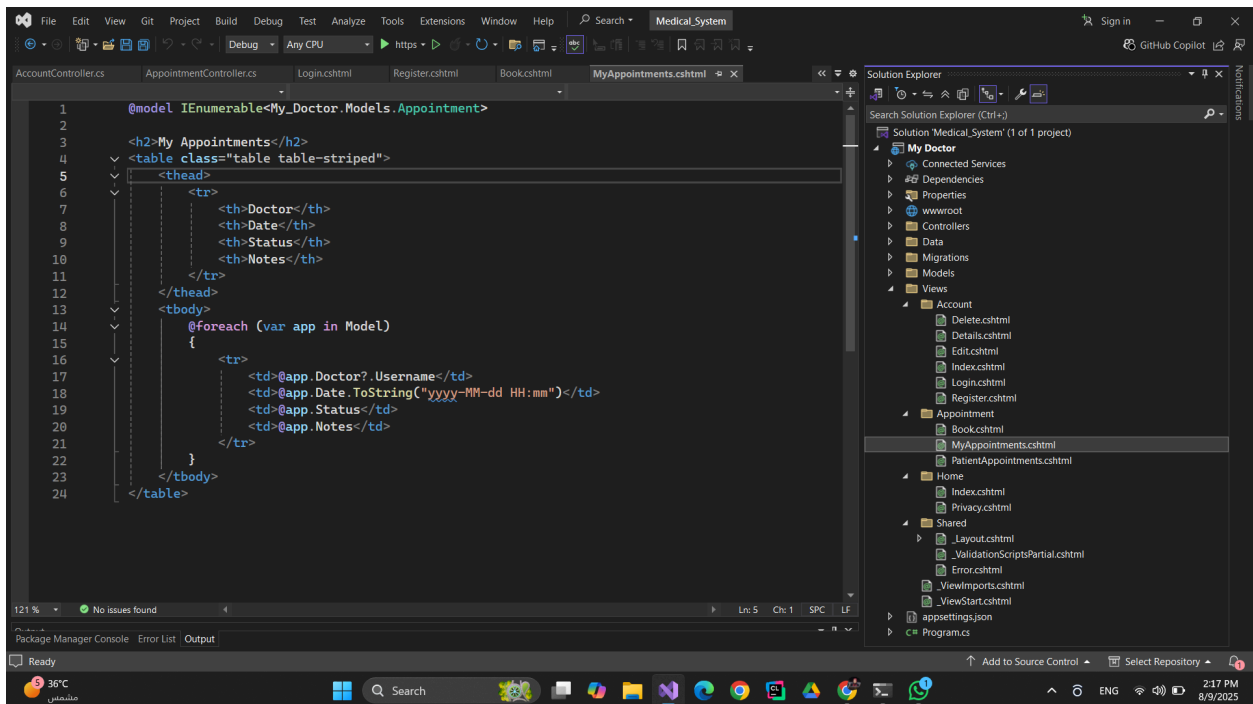


Oppointment :

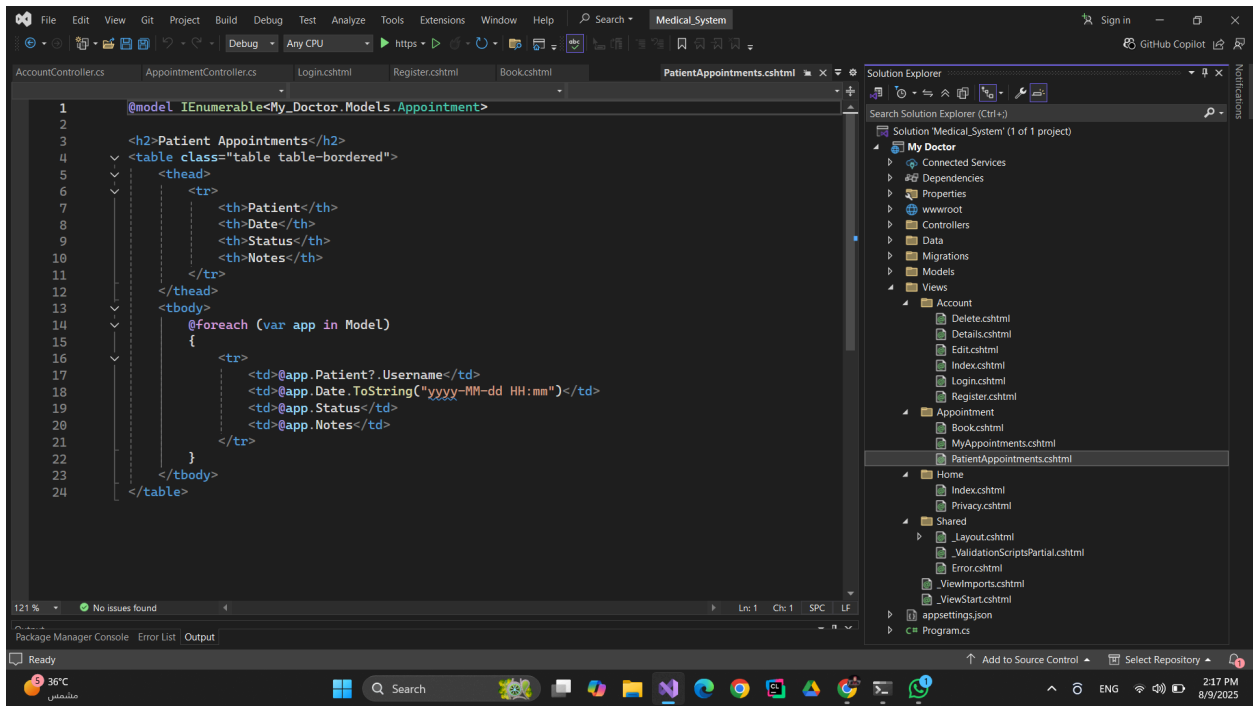
Book : Form for a patient to select a doctor, choose a date, and request an appointment.



MyAppointments : Displays a list of all appointments booked by the logged-in patient.

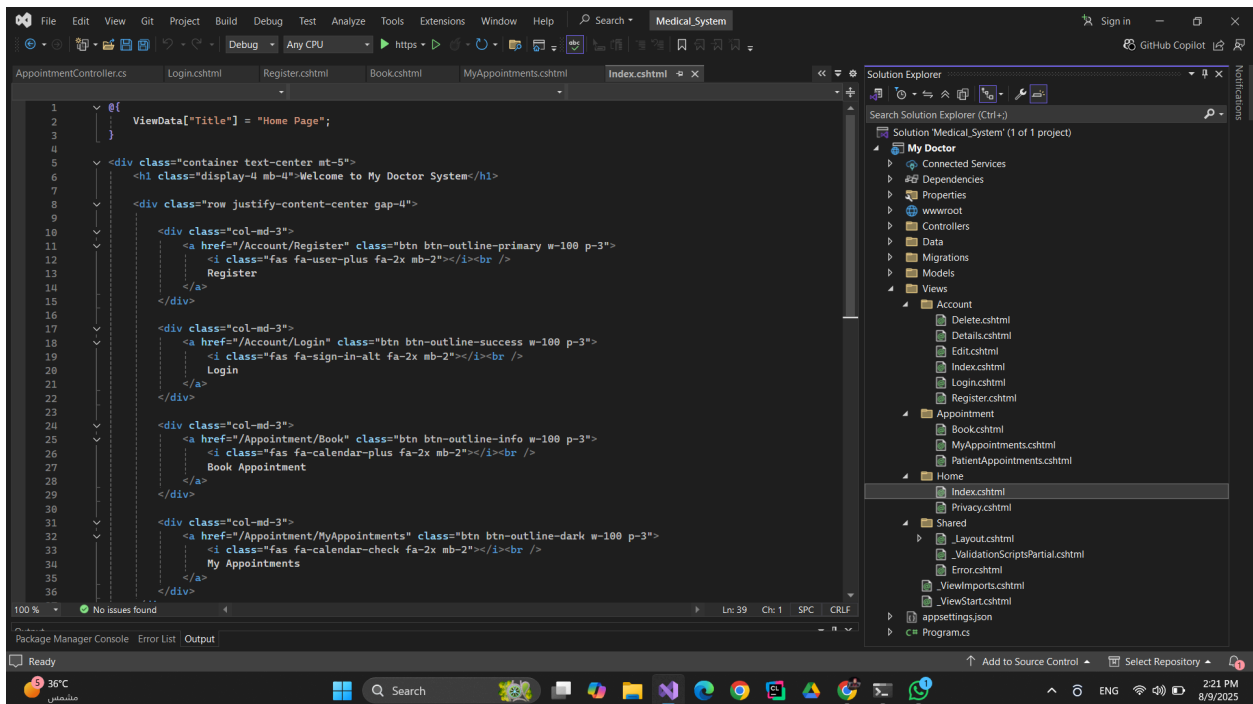


PatientAppointments : Displays all appointments for the logged-in doctor, showing their patients' details.



Home :

Index – The home page of the system with navigation to main features.



Privacy – Displays the application's privacy policy.

Error – Shows an error message with the request ID for debugging.